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Akshat Dhingra

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ABOUT ME

Recent BSc Computer Systems graduate (Riga Technical University) specialising in High-Performance Computing (HPC) and large-scale data infrastructure. My technical background bridges advanced computing and real-world application, demonstrated through a university-industry collaboration with SEB Bank, where I leveraged HPC environments to process high-volume datasets and deploy scalable predictive models. Currently, I am expanding my academic research through an ongoing collaboration with Professor Jordi Cabot from the University of Luxembourg, focusing on software engineering and DevOps automation. Driven by complex computational challenges, I am highly motivated to deepen my expertise in advanced parallel architectures by pursuing a rigorous master's programme in HPC.

EDUCATION AND TRAINING

19/01/2022 – 19/01/2026 Riga, Latvia

BSC. COMPUTER SYSTEMS Riga Technical University

Level in EQF EQF level 6

15/02/2025 – 15/07/2025 Riga, Latvia

HIGH-PERFORMANCE COMPUTING PROGRAMME Riga Technical University

Level in EQF EQF level 7

WORK EXPERIENCE

HPC SOLUTIONS DEVELOPER – SEB BANK (UNIVERSITY–INDUSTRY COLLABORATION) – 15/02/2025 – 15/07/2025 – RIGA, LATVIA

Leveraged High-Performance Computing (HPC) infrastructure to architect and execute highly efficient, large-scale data processing pipelines for advanced financial analytics.
Engineered computationally intensive predictive models (including SVM and XGBoost regression) on HPC clusters to analyze massive historical financial datasets and forecast cash flow trends.
Developed a scalable machine learning classification system to segment corporate clients into distinct financial behavior categories (Lumpy, Seasonal, Stable, Negative, Steady Growth) for large-scale risk assessment.
Optimized data integration and preprocessing workflows to handle the high volume and velocity of SEB Bank's Big Data requirements.
Presented the end-to-end computational methodology, AI-driven insights, and technical results to domain experts and stakeholders at SEB Bank and Riga Technical University.

APPLIED AI & HPC SOLUTIONS DEVELOPER – TOMEGA.LV – 12/02/2026 – Current – RIGA, LATVIA

Architecting the High-Performance Computing (HPC) backend for an AI-driven educational technology solution with Tomega, designed to eliminate "technology anxiety" by simplifying the operation of complex classroom and office hardware.
Leveraging university HPC clusters for large-scale model training, utilizing multi-node environments to process massive datasets of device telemetry, technical manuals, and user interaction logs.
Implementing distributed computing pipelines to accelerate the fine-tuning of machine learning algorithms, ensuring the AI can quickly and accurately troubleshoot projectors, smart boards, and printers in real-time.
Bridging the gap between heavy computational infrastructure and seamless user experience, translating scalable backend architectures into a low-latency, intuitive interface for non-technical users.

RESEARCH ASSISTANT – RIGA TECHNICAL UNIVERSITY – 25/11/2025 – Current – RIGA, LATVIA

Co-conducting applied research on Model-Driven DevOps with AI (MDDOAI) in direct collaboration with Prof. Jordi Cabot (University of Luxembourg), focusing on the automated generation of CI/CD pipelines from complex software architecture models.
Architected and implemented scalable DevOps workflows, utilizing advanced model-to-model (M2M) and model-to-text (M2T) transformation techniques to automate infrastructure deployment and continuous integration.
Deployed and configured the MDDOAI framework to execute end-to-end automated transformations, converting platform-independent system models directly into executable GitLab CI/CD pipelines.
Analyzed architecture-driven pipeline models to optimize build, test, and deployment processes, a critical skill for managing large-scale, automated computational environments.
Conducted systematic literature reviews in Model-Driven Software Engineering (MDSE) and AI-driven automation, contributing to methodology refinement for next-generation infrastructure deployment.

DATA SOFTWARE ENG. TRAINEE – EPAM – 15/03/2026 – Current – RIGA, LATVIA

Selected from a highly competitive applicant pool for an intensive, enterprise-grade data software engineering program focused on scalable data architectures and distributed computing.
Architecting and optimizing high-throughput ETL/ELT pipelines, mastering the end-to-end lifecycle of big data from ingestion and storage to processing and analytics.
Gaining hands-on expertise in modern data engineering stacks, utilizing distributed processing frameworks (such as Apache Spark/Hadoop), messaging queues (e.g., Kafka), and scalable SQL/NoSQL databases to manage massive data volumes.
Applying rigorous software engineering best practices to data infrastructure, integrating version control, CI/CD, and cloud computing principles that directly complement advanced High-Performance Computing workflows.

WEB DEVELOPER – FORMULA STUDENT RTU – 01/10/2025 – Current – RIGA, LATVIA

Served as the sole web developer, taking full ownership of **building** the official Formula Student RTU website from the ground up and **managing** its ongoing technical infrastructure and performance.
Bridged the gap between technical engineering teams and external stakeholders by translating complex automotive development milestones into accessible, engaging digital content.
Managed digital communications and outreach strategies to drive project visibility, enhance sponsor engagement, and support funding initiatives.
Collaborated within a fast-paced, multidisciplinary student engineering team, effectively balancing development tasks and communication efforts under strict international competition deadlines.

GRAPHIC DESIGNER – INVENTIVE CAFE INDIA PVT. LTD. – 15/01/2021 – 15/06/2021 – VIJAYAWADA, INDIA

Initiated a career trajectory in the IT and digital product space, starting with the creation of foundational brand assets (ranging from corporate logos to digital web graphics) that sparked a broader interest in web infrastructure and technology.
Designed and delivered digital visual content across various platforms, developing an early understanding of user-facing digital environments and visual communication.
Produced high-quality graphics utilizing industry-standard design tools (Adobe Photoshop, Illustrator) while strictly aligning with corporate brand guidelines and project requirements.
Collaborated with cross-functional teams to ensure visual design initiatives effectively supported broader digital communication and outreach strategies.

HONOURS AND AWARDS

28/01/2026
3rd Place (Bronze) – LG AI Hackathon – LG

Engineered a fully functional, product-ready Smart Home Hub that utilizes Artificial Intelligence to optimize household energy consumption and reduce electricity waste.
Developed real-time IoT data pipelines to process incoming sensor data and execute predictive machine learning models for intelligent resource allocation.
Recognized by LG technical judges for architectural scalability, commercial viability, and the successful deployment of AI in an edge-computing environment.
Collaborated in a rapid-development environment, demonstrating strong cross-functional teamwork and system design skills under a strict deadline to secure a top-3 podium finish.

PROJECTS

15/03/2025 – 21/01/2026

Methods in the Detection of Known and Unknown Fraud Patterns Using Human-Interpretable Feature Engineering

Architected a scalable data processing pipeline to ingest and transform the massive IEEE-CIS Fraud dataset, engineering human-interpretable behavioral features (time-window aggregates) while utilizing time-aware splitting and SMOTE to manage severe class imbalance.
Engineered a computationally intensive hybrid stacking ensemble, training diverse base models (LightGBM, Isolation Forests, and Deep Autoencoders) to capture both established fraud topologies and zero-day (unknown) anomalies.
Optimized algorithmic performance via distributed hyperparameter tuning using Optuna, managing the computational overhead of the ensemble architecture to maximize precision, recall, and AUC-PR metrics.
Deployed an XGBoost meta-learner integrated with Explainable AI (SHAP), successfully bridging the gap between high-performance, high-dimensional machine learning and transparent, human-readable analytical outputs for fraud investigators.

SKILLS

Data analysis and analytics

Data analysis | Analytical thinking | Data-driven decision making | Financial data analysis | Data interpretation

Programming and technical skills

Python programming | SQL (Structured Query Language) | CICD Tools (Jenkins/Gitlab CI)

Machine learning and statistics

Machine learning | Predictive analytics | Statistical analysis | Classification and regression | Handling imbalanced data

Data processing and engineering

Data preprocessing | Data cleaning | Data integration | Large-scale data processing | High-Performance Computing (HPC)

Communication and collaboration

Stakeholder communication | Team collaboration | Technical documentation | Presentation of analytical results

High-Performance Computing & Infrastructure

linux shell scripting | Containerization (Docker, kubernetes) | CUDA (GPU) | Parallel programing

LANGUAGE SKILLS

Mother tongue(s): **HINDI**

Other language(s): **ENGLISH** | **GERMAN**